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Should vibe-coding serve as the primary or unsupervised driver in professional software development workflows?

I. INTRODUCTION

According to NVIDIA CEO Jensen Huang (Feb 2024), traditional coding is becoming obsolete quoting “coding is dead” as a future skill. He advises people to skip programming languages and focus on fields like biology or manufacturing, since AI will handle code from natural language descriptions, so “nobody has to program anymore.” However, vibe-coding should not drive professional workflows. Human thinking, judgment, and oversight remain essential to prevent risks, ensure quality, and avoid technical debt or failures. When used responsibly as a supporting tool, it boosts speed, lowers barriers, raises productivity, improves code quality, and aids learning.

II. BODY

Primary or unsupervised vibe-coding in professional work creates severe technical debt through security holes and weak designs, making long-term maintenance and reliability impossible without human control. AI code often includes insecure patterns and ignores overall system architecture, which builds up costly fixes over time.

First, security risks stand out clearly. For example, Veracode’s 2025 GenAI Code Security Report found that 45% of AI-generated code introduced OWASP Top-10 vulnerabilities across tests on over 100 models and common languages like Java, Python, C#, and JavaScript. Similarly, a 2023 Stanford study showed that developers using AI assistants wrote less secure code while overestimating its safety. In addition, GitClear’s 2025 analysis of millions of lines revealed a 4× rise in code clones and higher short-term churn due to inconsistent AI outputs, further worsening maintainability.

Beyond security, this approach produces fragile, unscalable systems with poor patterns, bad performance, and missed edge cases that grow worse in production. Consequently, technical debt and brittleness become major problems. For instance, Ox Security’s 2025 report described AI code as “highly functional but lacking architectural judgment,” leading to bloat and scalability issues. Likewise, Sonar’s 2025 developer survey found that 53% of developers saw AI increasing technical debt through unreliable but seemingly correct code. Moreover, a 2025 MIT Sloan analysis warned that unchecked AI adds debt that harms scalability and system stability.

Recent real-world incidents reinforce these concerns. In 2025, flaws in tools such as Amazon Q, Cursor, GitHub Copilot, and others allowed prompt injections, data leaks, and remote code execution (for example, “IDESaster” research identified over 30 vulnerabilities across platforms).

Critics may argue that AI-reliant developers can still enter professional environments through freelancing, startups, or open-source contributions, potentially introducing vulnerable code into production systems. However, companies are raising standards and requiring stronger skills in architectural oversight, threat modeling, security audits, and human-AI collaboration to ensure that overreliance on AI does not compromise professional workflows.

III. CONCLUSION

In conclusion, vibe coding should not serve as the main or independent driver in professional software development. Although AI tools can improve speed and make coding more accessible, they cannot replace human critical thinking, responsible decision making, and the ability to ensure software quality and safety.

To move forward responsibly, developers and organizations should use vibe coding only as a supporting tool, not as a replacement for human expertise. Human oversight must remain present in planning, reviewing, and testing software to prevent security risks, technical debt, and system failures. This approach is more acceptable because it balances innovation with accountability and protects users from avoidable errors.

Professional software development requires careful judgment, ethical responsibility, and long-term thinking—qualities that AI alone cannot provide. If technology is meant to help people, should it not always be guided by human wisdom and responsibility?

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